

CS 260: Foundations of Data Science

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Fall 2026



HAVERFORD
COLLEGE

Admin

- **Sit somewhere new**
- **Lab 3** due tonight at midnight
- **Lab 4** posted (due next Tuesday)

Outline for today

- Recap SGD (stochastic gradient descent)
- Introduction to classification
 - Decision tree models
 - Probabilistic interpretation
- Evaluation Metrics
 - Confusion matrices
 - Precision and recall
 - ROC curves

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Stochastic Gradient Descent for Linear Regression

Key Idea: take the derivative of **one datapoint** at a time and use that to update w

set $\vec{w} = \vec{0}$

while **cost** $J(\vec{w})$ still changing (or max iter not reached):

 shuffle data points

 for $i = 1 \dots n$:

$$\vec{w} \leftarrow \vec{w} - \alpha \nabla J(\vec{w})_{x_i}$$

 store $J(\vec{w})$

Mini-quiz for linear regression

For each of the following terms/descriptions, write out the corresponding equation:

- 1) Linear regression **model**
- 2) Linear regression **cost function**
- 3) **Gradient** of cost function wrt one datapoint
- 4) **Gradient descent** weight vector update

Mini-quiz for linear regression

$$1. h_{\vec{w}}(\vec{x}) = \vec{w} \cdot \vec{x} = \hat{y}$$

$$2. J(\vec{w}) = \frac{1}{2} \sum_{i=1}^n (\overset{\text{truth}}{\downarrow} y_i - \overset{\text{prediction}}{\nearrow} \vec{w} \cdot \vec{x}_i)^2$$

$$3. \nabla J_{\vec{x}_i}(\vec{w}) = (\vec{w} \cdot \vec{x}_i - y_i) \vec{x}_i$$

$$4. \vec{w} \leftarrow \vec{w} - \alpha \nabla J_{\vec{x}_i}(\vec{w})$$

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Binary classification examples

- Transactions that indicate credit card fraud
- Accounts that are bots
- Detecting which scans show tumors
- Prenatal test for Down's Syndrome
- Finding genes under natural selection
- Regions of the environment that contains the object the robot is searching for

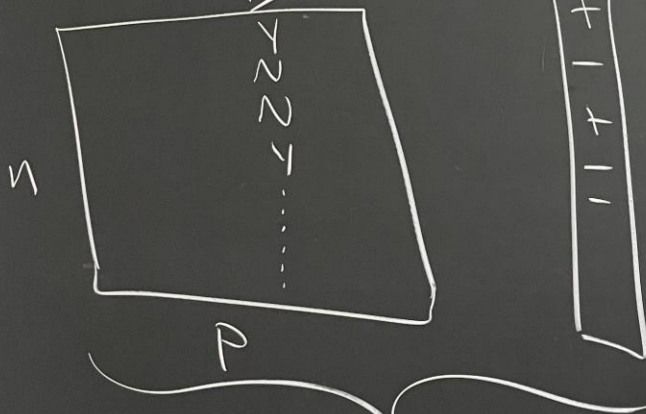
In all these examples, we are trying to find unusual items ("needle in a haystack") -- we call these *positives*

Introduction to Classification

Introduction to Classification

Single feature models $\vec{y}$

X_{fever}



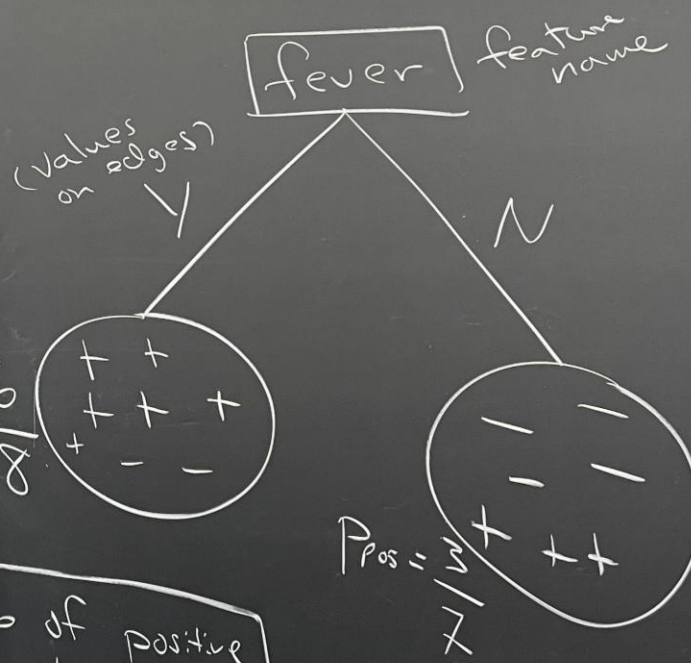
training data

+ disease
- no disease

$$P_{\text{pos}} = \frac{6}{8}$$

P_{pos} = prob of positive classification

model: decision tree with 1 feature ("stumps")



Introduction to Classification

New idea: use probabilities to classify test example

$$\vec{x}_{\text{test}} = [\cdots \overset{\text{fever}}{\downarrow} N \cdots]^T$$

$p_{pos} \geq \text{threshold} \rightarrow \text{classify as positive}$

Threshold = 0.5

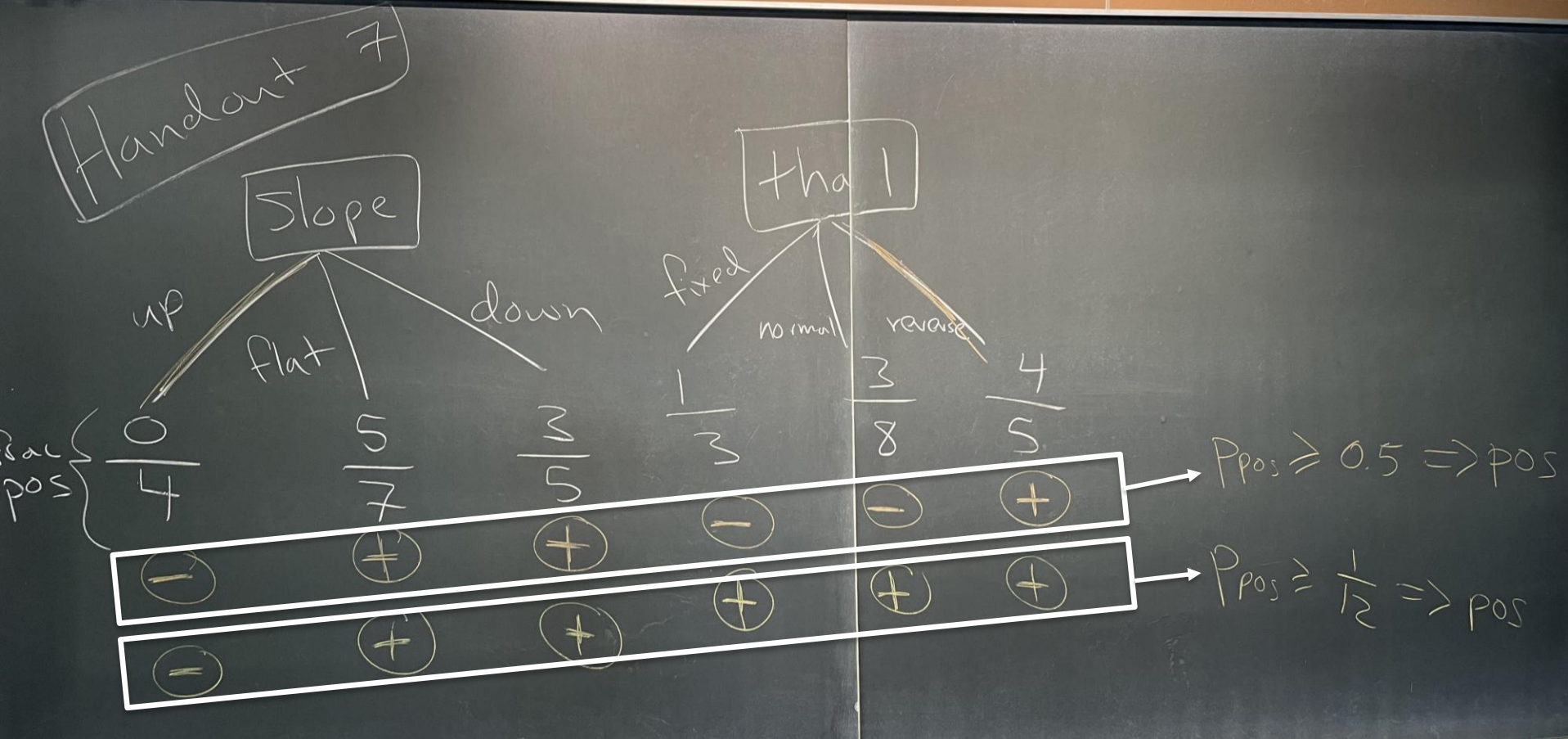
$\hat{y}_{\text{test}} = \text{negative (no disease)}$

Threshold = 0.25

$\hat{y}_{\text{test}} = \text{positive (disease)}$

Handout 7

Handout 7



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Goals of Evaluation

- Think about what metrics are important for the problem at hand
- Compare different methods or models on the same problem
- Common set of tools that other researchers/users can understand

Training and Testing

(high-level idea)

- **Separate** data into “**train**” and “**test**”
 - n = num training examples
 - m = num testing examples
- **Fit** (create) the model using **training data**
 - e.g., sea_ice_1979-2012.csv
- **Evaluate** the model using **testing data**
 - e.g., sea_ice_2013-2020.csv

Confusion Matrices

Test data:

80 negatives, 20 positives (m=100)

Confusion Matrices:

Note: all the same model,
different thresholds!

predicted

	-	+
true -	65	15
true +	7	13

Accuracy = $\frac{65+13}{100} = \frac{78}{100}$

	-	+
-	50	30
+	1	19

Accuracy = 0.69

Lower threshold = more
positive predictions

	-	+
-	76	4
+	11	9

Accuracy = 0.85

Higher threshold = more
negative predictions

$$\text{Accuracy} = \frac{\text{number of correct predictions}}{m} = \frac{1}{m} \sum_{i=1}^m \mathbb{1}(y_i = \hat{y}_i)$$

Confusion Matrices

		Predicted class	
		Negative	Positive
True class	Negative	True negative (TN)	False positive (FP)
	Positive	False negative (FN)	True positive (TP)

Confusion Matrices

Predicted class

Negative

Positive

Negative

True negative
(TN)

False positive
(FP)
“false alarm”

N (total number of true negatives)

True
class

Positive

False negative
(FN)
“miss”

True positive
(TP)

P (total number of true positives)

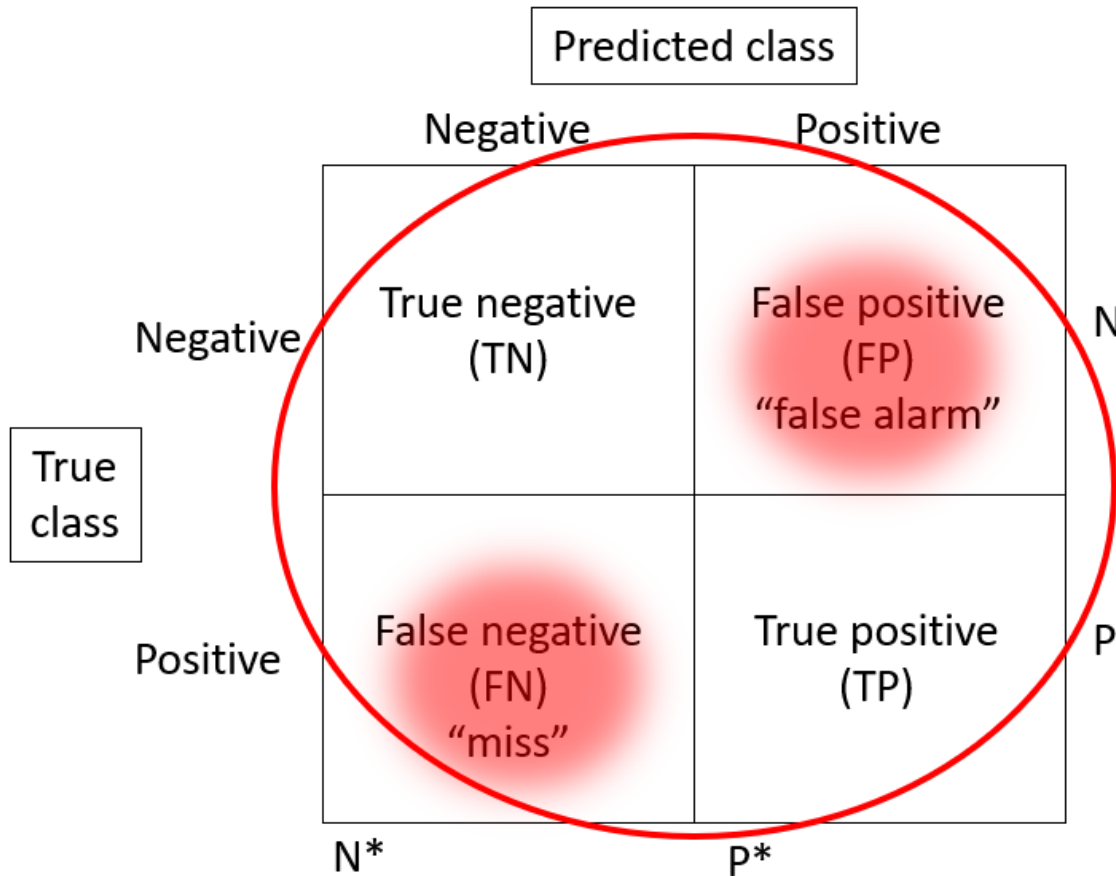
N* (what we said
was negative)

P* (what we said was
positive “flagged”)

Confusion Matrices

		Predicted class		
		Negative	Positive	
True class	Negative	True negative (TN) ✓	False positive (FP) “false alarm” ✗	N
	Positive	False negative (FN) “miss” ✗	True positive (TP) ✓	p
		N*	p*	

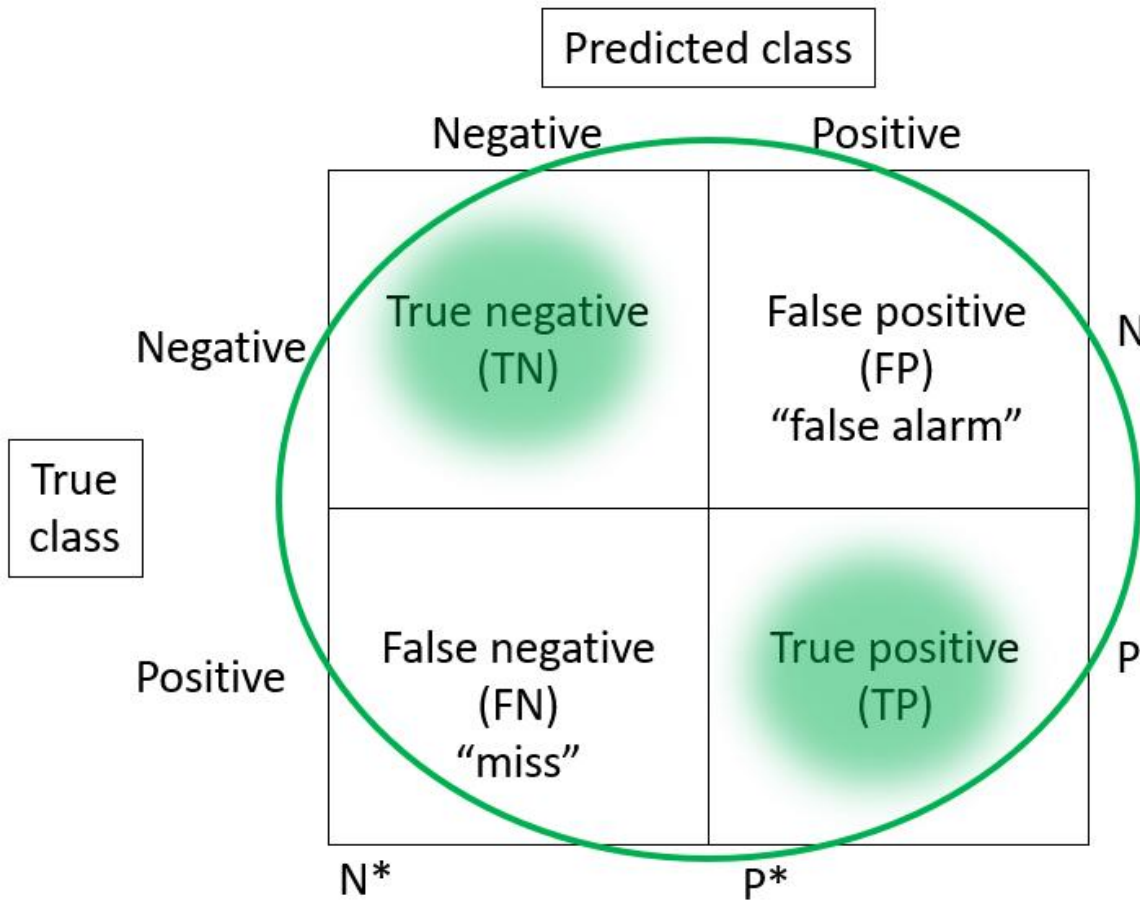
Confusion Matrices



Error:

$$\frac{(FN + FP)}{(TN + FP + FN + TP)}$$
$$= \frac{(FN + FP)}{(N + P)}$$

Confusion Matrices



Accuracy = 1-Error:

$$\frac{(TN + TP)}{(TN + FP + FN + TP)}$$
$$= \frac{(TN + TP)}{(N + P)}$$

Confusion Matrices

		Predicted class		
		Negative	Positive	
True class	Negative	True negative (TN)	False positive (FP) "false alarm"	N
	Positive	False negative (FN) "miss"	True positive (TP)	P
		N*	p*	

Precision:

$$\frac{TP}{(FP + TP)} = \frac{TP}{P^*}$$

Confusion Matrices

		Predicted class		
		Negative	Positive	
True class	Negative	True negative (TN)	False positive (FP) “false alarm”	N
	Positive	False negative (FN) “miss”	True positive (TP)	P
		N*	p*	

Recall
(True Positive Rate):

$$\frac{TP}{(FN + TP)} = \frac{TP}{P}$$

Confusion Matrices

		Predicted class		
		Negative	Positive	
True class	Negative	True negative (TN)	False positive (FP) "false alarm"	N
	Positive	False negative (FN) "miss"	True positive (TP)	P
		N*	p*	

False Positive Rate:

$$\frac{FP}{(TN + FP)} = \frac{FP}{N}$$

Precision and Recall

- Precision: of all the “flagged” examples, which ones are actually relevant (i.e., positive)?

(Purity)

- Recall: of all the relevant results, which ones did I actually return?

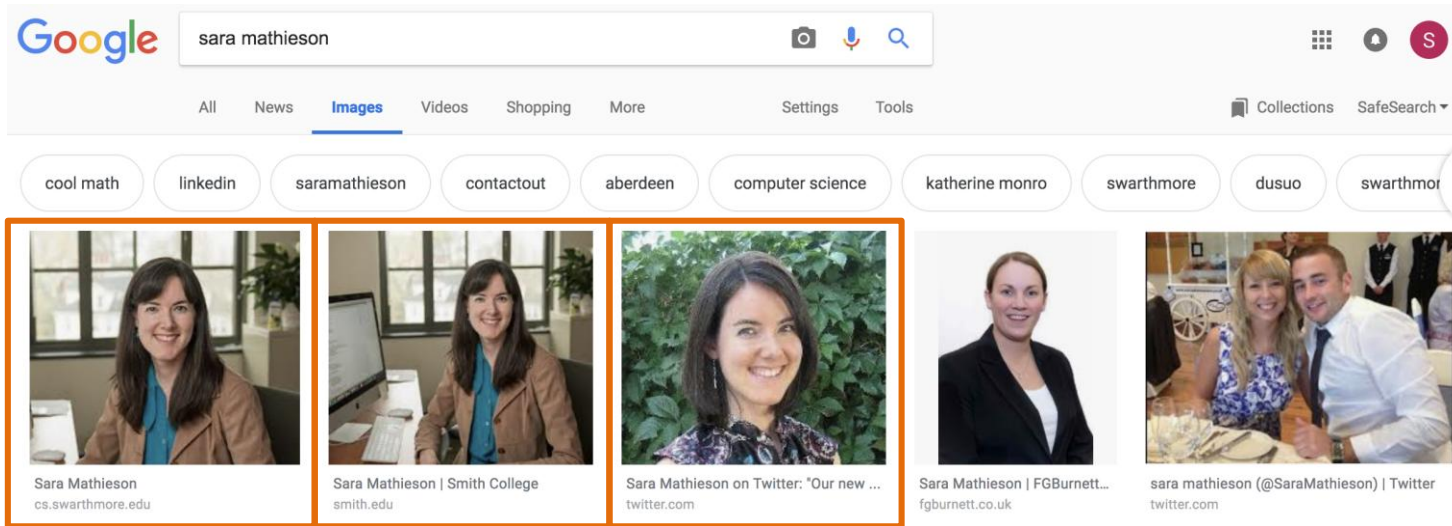
(Completeness)

Bernie

Me before
I took
the class

**I am once again asking
for high precision and high recall**

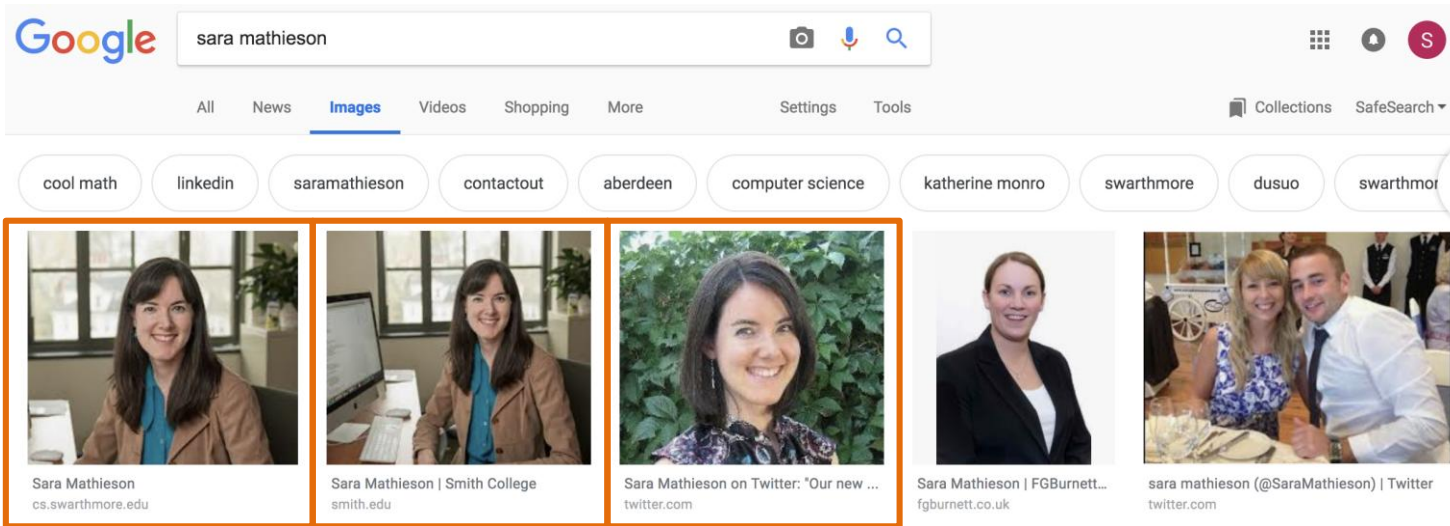
Precision and Recall



$P=6$ (number of images that are actually Sara)

- Precision?
- Recall?

Precision and Recall

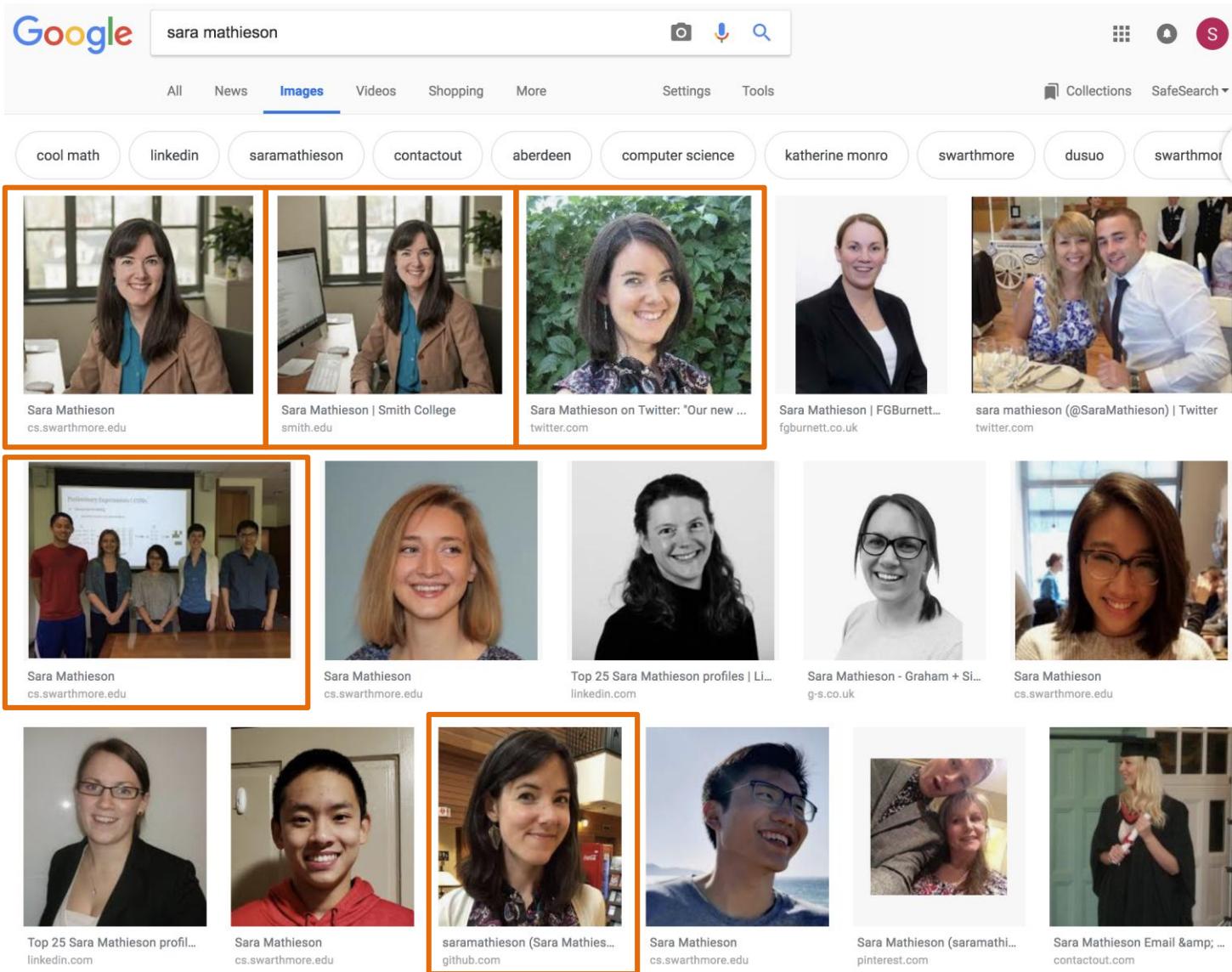


$P=6$ (number of images that are actually Sara)

- Precision = $\frac{TP}{(FP+TP)} = \frac{3}{5}$

- Recall = $\frac{TP}{(FN + TP)} = \frac{3}{6}$

Precision and Recall



$P=6$ (number of images that are actually Sara)

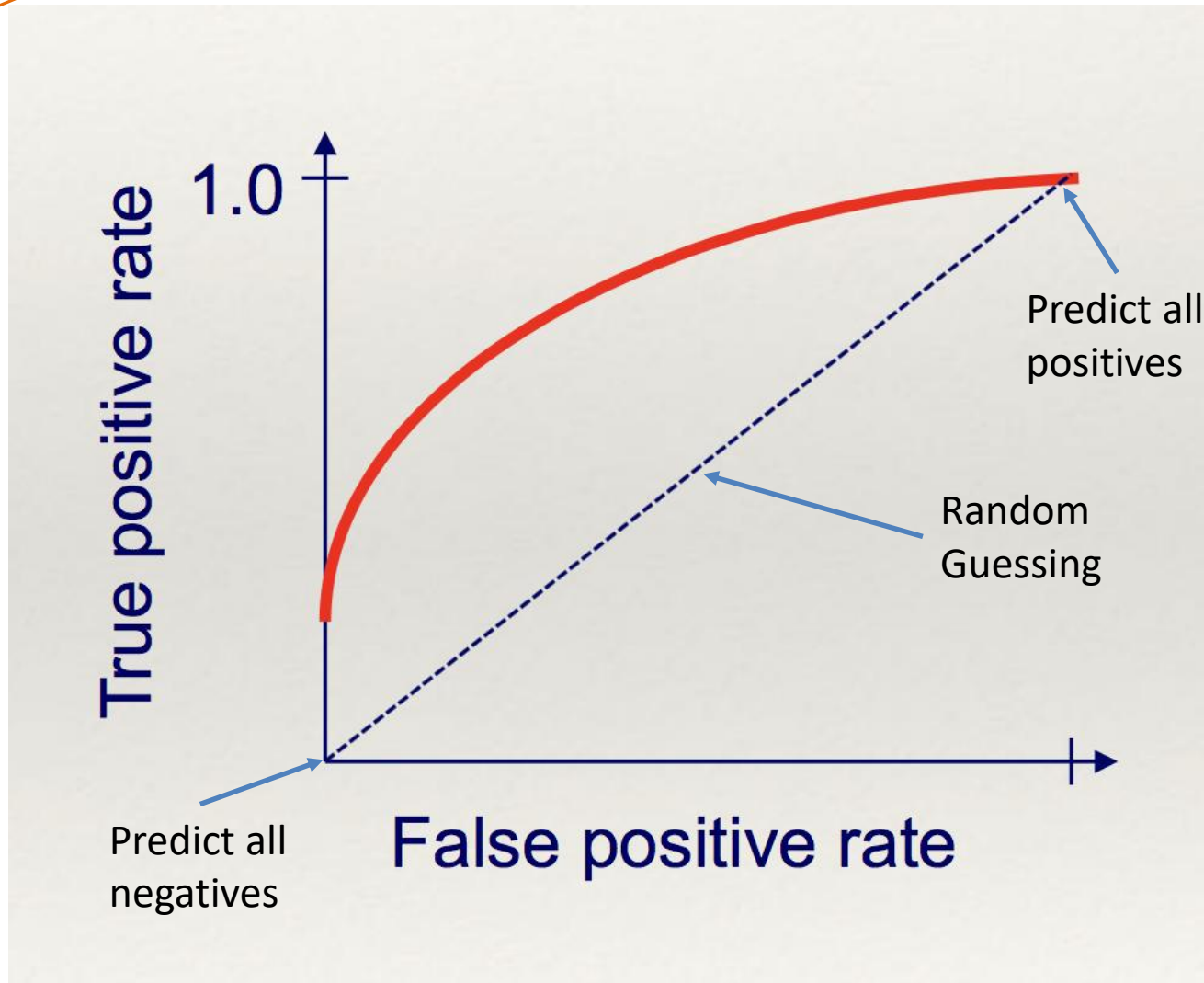
- Precision = $5/16$
- Recall = $5/6$

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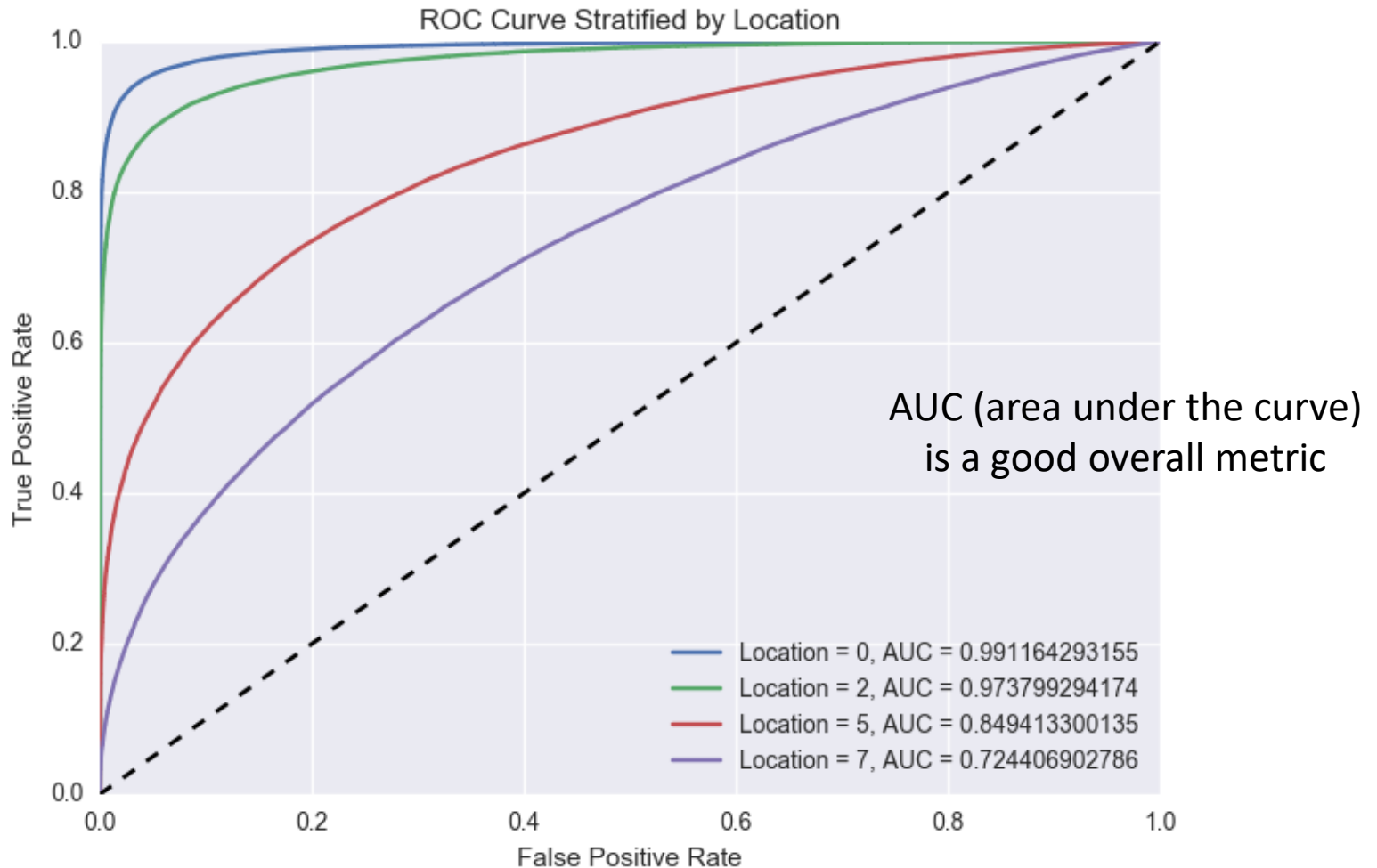
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ROC curve (Receiver Operating Characteristic)

More history here!



ROC curve example: comparing methods



Example of a ROC curve

Chan, Perrone, Spence, Jenkins, Mathieson, Song